Tanvir Ahmed — AI Engineer Knowledge Base

Professional Profile

Tanvir Ahmed is an AI Engineer / Machine Learning Engineer specializing in end-to-end AI systems across Machine Learning, Deep Learning, Computer Vision, Generative AI, Retrieval-Augmented Generation (RAG), and MLOps.

He has practical experience designing, developing, evaluating, deploying, and serving AI applications using Python, TensorFlow, PyTorch, LangChain, LangGraph, FastAPI, Docker, Pinecone, AWS, and related tools.

Core Technical Areas

Machine Learning

• Supervised and unsupervised learning

• Regression and classification

• Feature engineering and preprocessing

• Model evaluation and hyperparameter tuning

• Scikit-learn, XGBoost, CatBoost

• Pandas, NumPy, Matplotlib, Seaborn

• Explainability with SHAP

• Model deployment with FastAPI

Deep Learning

• TensorFlow and PyTorch

• Transfer learning

• CNN architectures

• VGG16

• ResNet50

• U-Net

• YOLO

• Segment Anything Model (SAM)

• Data augmentation

• Image classification, object detection, localization, and segmentation

Computer Vision

• OpenCV

• Medical image analysis

• MRI brain tumor analysis

• Chest X-ray analysis

• Image preprocessing and augmentation

• Object detection

• Semantic/instance-style segmentation workflows

• Tumor localization and mask visualization

Generative AI

• Large Language Models (LLMs)

• OpenAI API

• Gemini

• LangChain

• LangGraph

• Retrieval-Augmented Generation (RAG)

• Multimodal RAG

• Vision-language workflows

• Prompt engineering

• Structured LLM output

• Agentic and multi-agent systems

• CrewAI

RAG and Information Retrieval

• Document ingestion and parsing

• PDF text extraction

• Table extraction

• Embedded image extraction

• Image-to-text descriptions

• Chunking and metadata

• Semantic embeddings

• Vector search

• Pinecone

• ChromaDB

• Hybrid retrieval

• Retrieval evaluation

• Document relevance grading

• Query routing

• Context refinement

• Hallucination reduction

• Source-grounded answer generation

LangGraph Architecture

Tanvir has worked with state-based LangGraph workflows containing specialized nodes such as:

• Retrieval decision/router

• Document retrieval

• Per-document relevance evaluation

• Web search fallback

• Context refinement

• Answer generation

• Ambiguous-question handling

• Incorrect/rejected retrieval handling

• Conditional routing based on evaluator verdicts

A typical workflow is:

User Question → Retrieval Decision → Retrieve Documents → Evaluate Documents → Refine Context → Generate Answer.

If retrieved documents are insufficient or ambiguous, the workflow can route to web search or another recovery step.

MLOps and Deployment

• MLflow

• DVC

• DagsHub

• GitHub Actions

• CI/CD

• Docker

• FastAPI

• AWS EC2

• AWS S3

• GCP

• Linux

• Bash

• Git

Major Projects

Multi-Modal Retrieval-Augmented Generation (RAG) System

Role: AI Engineer

Technologies:

OpenAI Vision, Pinecone, Python, LangChain, AWS.

Capabilities:

• Processes unstructured PDF documents.

• Extracts text, tables, and embedded images.

• Preserves relationships between document content and visual elements.

• Uses vision-language models to describe or interpret images.

• Uses semantic vector search for retrieval.

• Supports multimodal reasoning over document content.

• Uses hybrid retrieval to improve retrieval quality.

• Designed to reduce hallucinations by grounding generation in retrieved context.

AI Medical Assistant — LLM & RAG Healthcare System

Role: AI Engineer

Technologies:

LangChain, OpenAI API, Gemini, Pinecone, FastAPI, Python.

Capabilities:

• Medical question-answering using a RAG architecture.

• Retrieves relevant information from medical datasets/documents.

• Uses embeddings and Pinecone semantic search.

• Generates contextual, evidence-backed responses.

• Uses prompt templates and safety guardrails.

• Designed to communicate complex medical information in accessible language.

• Exposes inference through a FastAPI REST API.

NeuroVision AI — Brain Tumor Detection, Segmentation & RAG Assistant

Role: Lead Developer

Technologies:

TensorFlow, VGG16, YOLO, SAM, FastAPI, Pinecone, AWS, OpenAI.

System pipeline:

MRI Image → Classification → Tumor Localization → Tumor Segmentation → Visualization.

Components:

• VGG16 for brain tumor image classification.

• YOLO for tumor localization/object detection.

• Segment Anything Model (SAM) for tumor boundary extraction/segmentation.

• FastAPI for inference serving.

• RAG chatbot for brain-tumor-related document question answering.

• Pinecone for vector retrieval.

• LLM for grounded response generation.

• AWS for deployment.

The RAG component can work with brain-tumor medical documents containing information about tumor types, symptoms, imaging findings, diagnosis, pathology, grading, molecular markers, treatment, prognosis, research findings, tables, and statistics.

PneumoScan-AI — Automated Chest X-ray Segmentation

Role: AI Developer

Technologies:

TensorFlow, U-Net, ResNet50, OpenCV.

Capabilities:

• Automated lung segmentation from chest X-rays.

• ResNet50 backbone with transfer learning.

• U-Net segmentation architecture.

• Data augmentation.

• Abnormal-region highlighting.

• Achieved Dice similarity above 0.80 across varying image conditions.

Shipment Price Prediction System

Role: ML Engineer

Technologies:

Python, NumPy, Pandas, Scikit-learn, XGBoost/CatBoost, FastAPI, Docker, AWS.

Capabilities:

• Predicts shipment/logistics costs.

• Performs data preprocessing and feature engineering.

• Handles numerical and categorical features.

• Uses model training and hyperparameter tuning.

• Evaluates model performance.

• Provides real-time prediction through FastAPI.

• Containerized with Docker.

• Deployed using AWS infrastructure.

• Uses ML engineering practices for reproducible deployment.

Query Routing Guidance

When answering questions about Tanvir Ahmed, use this knowledge base as the primary source.

For questions about:

• Resume or professional profile → use the Professional Profile.

• AI/ML skills → use Core Technical Areas.

• RAG/LangGraph → use RAG and LangGraph sections.

• Brain tumor AI → use NeuroVision AI.

• Medical AI → use AI Medical Assistant and NeuroVision AI.

• Computer vision → use Deep Learning and Computer Vision sections.

• Deployment/MLOps → use MLOps and Deployment.

• Project-specific technologies → use the corresponding project section.

Do not invent:

• Employers not listed here.

• Job titles not listed here.

• Academic degrees beyond the listed BSc.

• Project metrics not explicitly stated here.

• Certifications, publications, awards, or work history not provided here.

• Specific model accuracy values unless explicitly stated.

• Specific datasets unless explicitly stated.

• Personal information not contained in this knowledge base.

If the answer is not supported by this knowledge base, say that the information is not available in the provided documents.

Education

Bachelor of Science (BSc)

Narsingdi Government College, Bangladesh.

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Professional focus: AI Engineering, Machine Learning, Deep Learning, Computer Vision, Generative AI, RAG, and MLOps.